Frito Lay Employee Attrition Analysis

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Overview

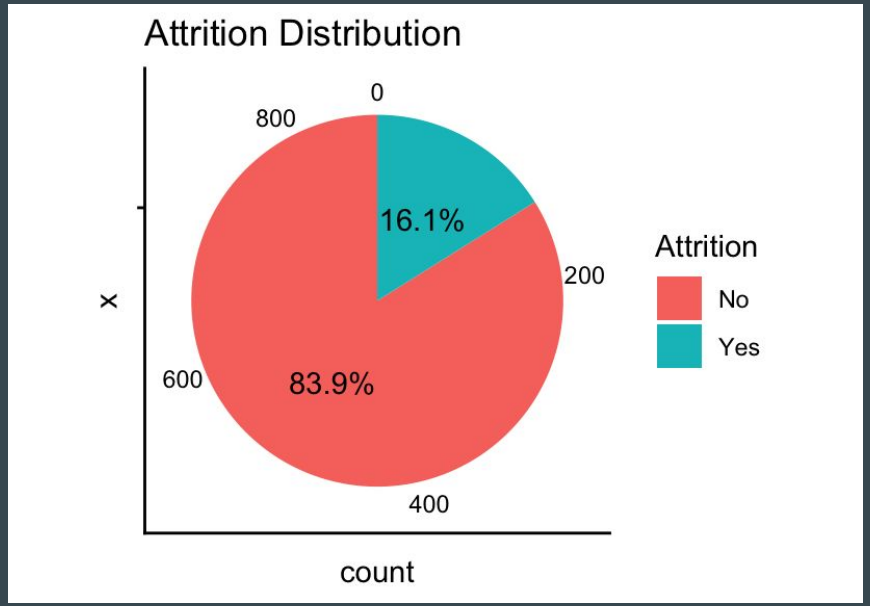
- DDSAnalytics was hired by Frito Lay to identify factors driving employee attrition
- Dataset: 870 employees, 35 variables
- Goal: Identify the top factors related to attrition and build models to predict it
- Business Impact: Replacing an employee costs 50-400% of their annual salary
- Frito Lay estimates $200 per employee in retention incentives

1. Exploring our data

Attrition Rate Distribution

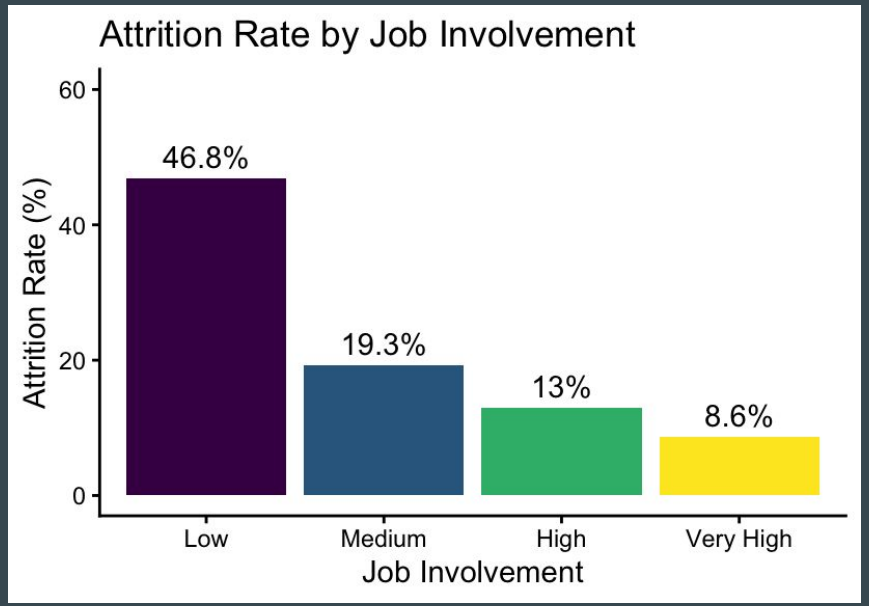
Total Employees: 1470

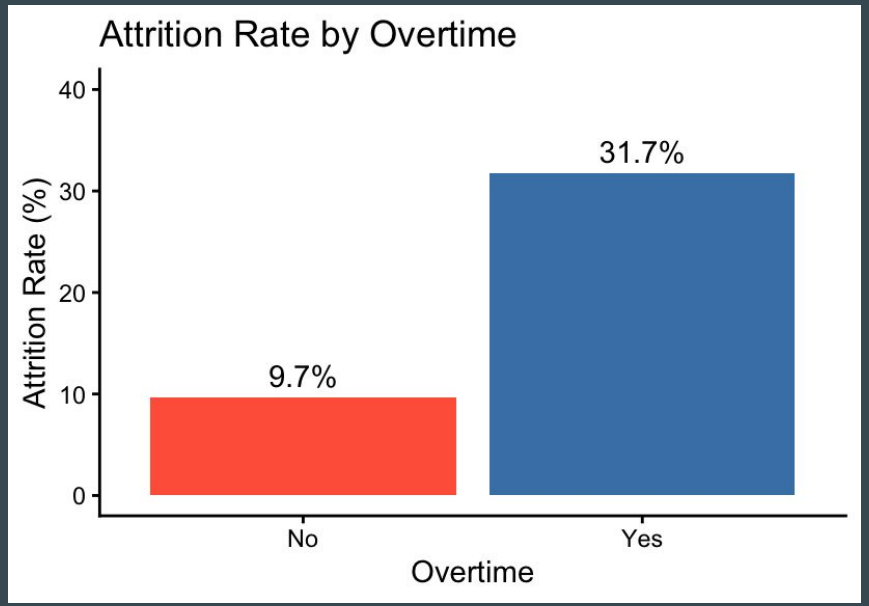
Attrition: 237 (16%)
No Attrition 1233 (84%)



Good Predictors of Attrition

Categorical: Job Involvement and Overtime

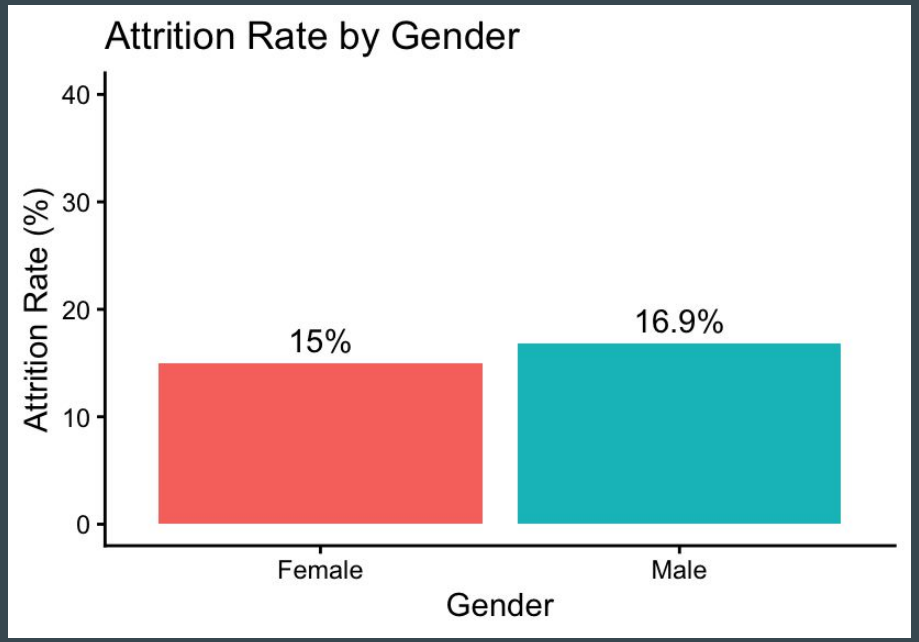


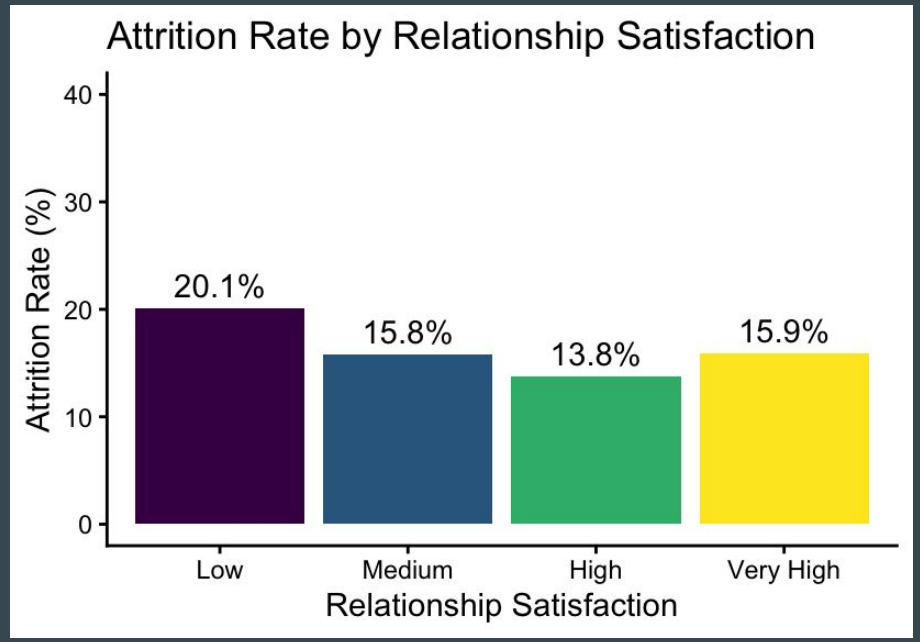


Continuous: Monthly Income and Total Working Years

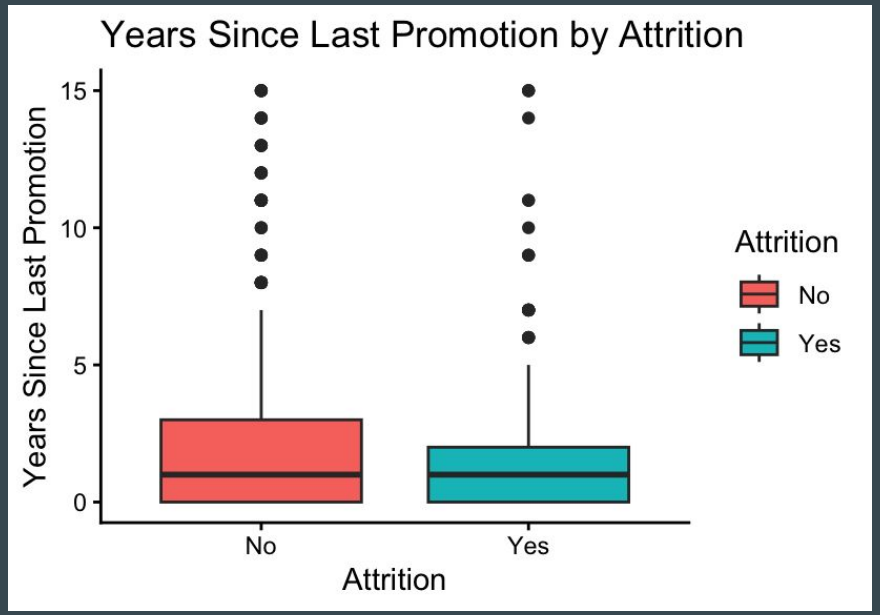
Bad Predictors of Attrition

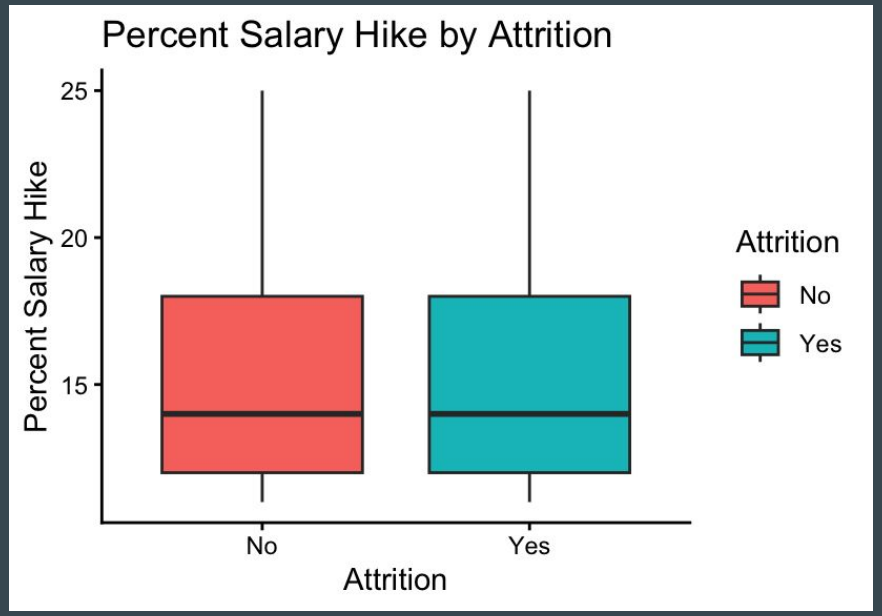
Categorical: Gender and Relationship Satisfaction





Continuous: Years Since Last Promotion and Percent Salary Hike



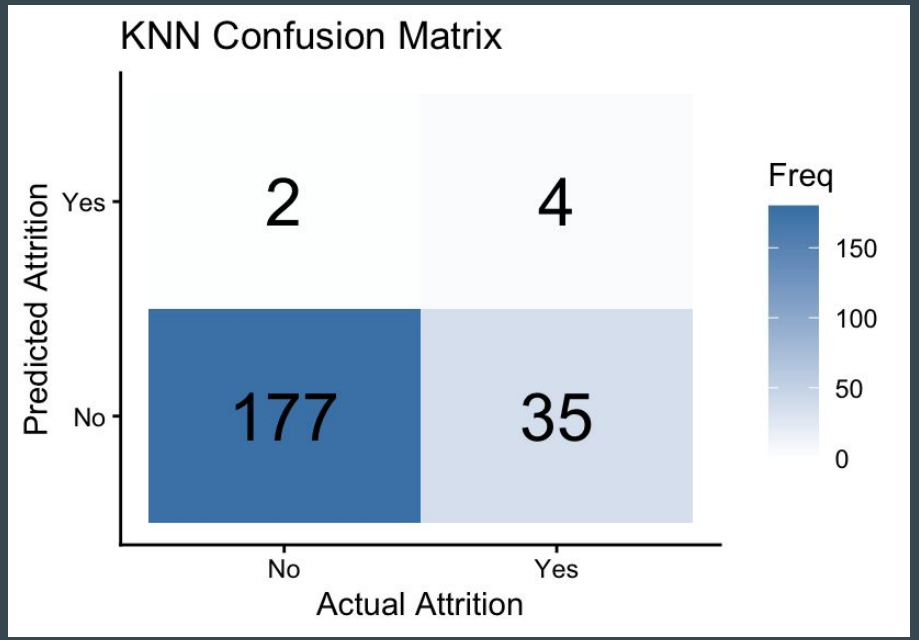


2. Building Our Models

Accuracy
Sensitivity
Specificity

KNN (K Nearest Neighbors)

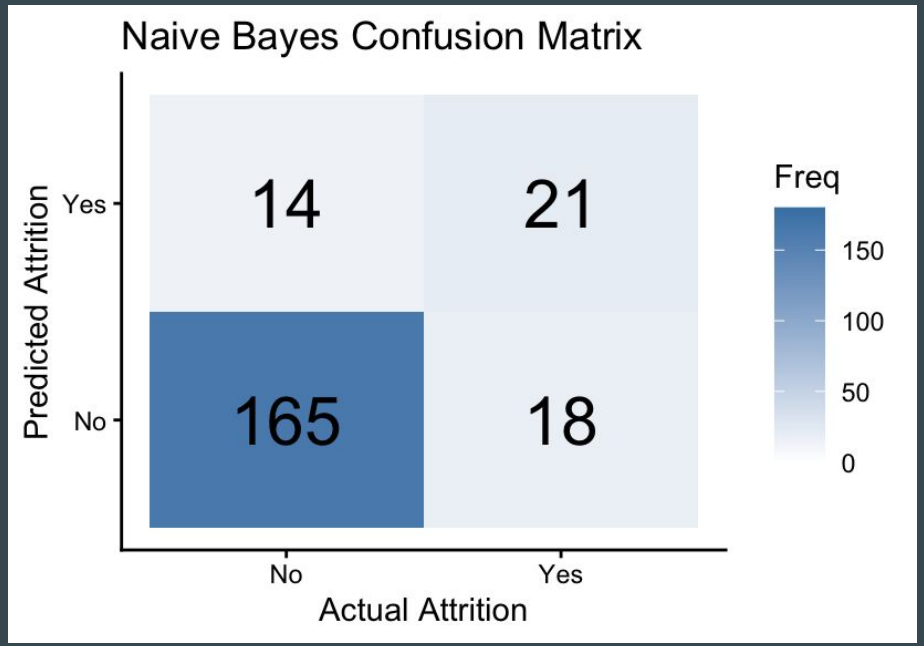
- Total Working Years
- Years in Current Role
- Monthly Income
- Age
- Years with Current Manager



Accuracy: .8303
Sensitivity: .6667
Specificity: .8349

Naive Bayes

- Total Working Years
- Years in Current Role
- Monthly Income
- Age
- Overtime
- Stock Option Level
- Job Involvement
- Marital Status



Accuracy: .8532
Sensitivity: .6000
Specificity: .90164

3. Model Savings

KNN vs. Naive Bayes

KNN:

Low End: $152,166

High End: $1,225,728

Naive Bayes:

Low End: $798,172

High End: $6,434,372

Conclusions

Top Findings:

- Employees working overtime are 3x more likely to leave
- Lower job involvement strongly predicts attrition
- Lower monthly income and fewer total working years signal higher risk

Our Recommendation:

- Use the Naive Bayes model to flag at-risk employees early
- Target retention incentives at overtime workers and low-involvement employees
- At just $200 per employee, proactive intervention can save Frito Lay up to $6.4 million

Questions?